

Chapter 7

Monetary and fiscal policy coordination

D. Andolfatto
Miami Herbert Business School
University of Miami

1 Introduction

Fiscal policy refers to the tax and spend policies put in place by our elected representatives. Congress determines government spending on goods and services, G_t . Congress also passes tax laws that interact with individual decisions to determine *net* tax revenue, T_t (by *net*, I mean net of transfers).¹ If the government was forced to balance its budget on a period-by-period basis, then $T_t = G_t$ at every date t .

In general, governments do not balance revenue and spending on a period-by-period basis. Typically, though not always, the primary surplus $T_t - G_t$ is negative; which is to say, the primary deficit $G_t - T_t$ is positive. It is the job of the U.S. Treasury to make sure the federal government can pay its bills. In the case of a primary deficit, the Treasury issues an array of securities to cover the shortfall. Historically, some of these securities took the form of zero-interest paper notes that served as money (recall Figure 5.1). Today, the U.S. Treasury mainly issues securities (USTs) that constitute promises of future money.² The stock of outstanding securities is labeled the national debt, D_t . If we let R_t denote the average gross nominal interest rate paid on USTs, then the interest expense of the national debt at date t is given by $(R_{t-1} - 1)D_{t-1}$. The total deficit in period t is given by the primary deficit

¹Transfers can be thought of as negative taxes.

²As explained in Chapter 1, USTs are converted into money usable at the retail level whenever they are purchased by a bank (central or private).

plus interest expense, $G_t - T_t + (R_{t-1} - 1)D_{t-1}$. The total deficit represents the change in the national debt, $D_t - D_{t-1}$; i.e.,

$$G_t - T_t + (R_{t-1} - 1)D_{t-1} = D_t - D_{t-1} \quad (1)$$

If the total deficit is positive, then the national debt is growing: $D_t > D_{t-1}$. If the total deficit is negative, then the national debt is shrinking: $D_t < D_{t-1}$. In this manner, we see that the tax and spend policies of the government also imply a corresponding debt management policy.

What is “monetary policy” in this context? Monetary policy essentially boils down to interest rate policy; i.e., determining how R_t should be set in a manner that helps achieve the central bank’s objectives. The yield on government debt can set either directly or indirectly by the central bank. Let me take a moment to explain this distinction.

The national debt can be divided into two components: $D_t = M_t + B_t$, where M_t denotes USTs held by the Fed and B_t denotes USTs held outside the Fed. M_t is called the *monetary base*. Historically, the monetary base consisted primarily of currency. Today, it consists largely of interest-bearing reserves. Let R_t^m denote the gross interest paid on Fed liabilities. Let R_t^b denote the gross interest rate paid on USTs held outside the Fed. Prior to 2008, $R^m = 1$. How did the Fed influence R_t^b when it was legally prohibited from paying interest on its liabilities? The answer is by changing the quantity of zero-interest reserves M_t through “open-market operations” or OMOs. So, for example, if the Fed wanted to drive down the market interest rate on bonds R_t^b , it would instruct the trading desk at the Federal Reserve Bank of New York to purchase USTs. These purchases would be financed by increasing M_t . The effect of such an OMO then is to increase M_t and decrease B_t , leaving D_t unchanged. Because interest-bearing USTs in the hands of private sector agents would now be lower, the price of USTs would be bid up—i.e., their yield, R_t^b , would decline.

In what follows, I simplify by assuming that M_t and B_t are perfect substitutes. This is not exactly true in reality, but the distinction does not matter for the results I derive below. What a central bank might achieve in the way of interest rate policy through OMOs that alter the ratio M_t/B_t can also be achieved by targeting money market interest rates directly. For simplicity then, let us just assume that monetary policy chooses R_t .

Equation (1) suggests that monetary and fiscal policy decisions must

somehow be reconciled with each other. There is no such thing as an interest rate policy that does not have fiscal implications. First, the policy interest rate influences the interest expense of the national debt. Second, interest rate policy can affect economic activity in a manner that influences net tax revenue and the trajectory of the debt.³ By the same token, fiscal policy cannot be implemented without reference to monetary policy. The fact that monetary and fiscal policy have to be consistent with each other follows from the arithmetic in (1). How monetary and fiscal policies are coordinated with each other is a different matter entirely. The economic implications of monetary and fiscal policy turn out to depend on how they are coordinated.

2 Model

2.1 Individual choices

Let us consider the model from the previous chapter with a slight variation designed to make the analysis simpler. The young have preferences

$$U(c_1, c_2) = u(c_1) + \beta c_2 \quad (2)$$

Assuming that utility is linear in future consumption implies that desired saving will not depend on future net income. Introducing the parameter β is a way to capture the idea that desired saving will depend on factors not related to interest rates or income. We can write the desired saving function for a young person as $s^D(r; \beta)$. Recall that $r = R/\Pi$. For these preferences:

[A1] $s^D(r; \beta)$ is increasing in r and in β .

The first property should be familiar to you—it is simply the substitution effect of an increase in r dominating any potentially offsetting wealth effect. An increase in β here literally reflects an increase in patience on the part of young individuals. However, I also want to think of β as a reduced-form way of capturing other forces that influence the demand for real money balances.

³For example, tax revenue typically declines in a recession. At the same time, transfers typically increase during a recession. Both of these effects serve to reduce T_t , increasing the primary deficit.

An increase in β , for example, may be the consequence of a regulation requiring banks to hold larger quantities of safe assets like U.S. Treasury securities.

2.2 Government budget constraint

The government inherits an initial nominal debt D_0 (owned by the initial old in the model). The government spends G_t dollars on goods and services in period t . The government collects T_t dollars in tax revenue in period t . To rationalize the desired saving function in [A1], assume that all tax revenue is collected as a lump-sum tax on the old.⁴ And to make absolutely clear that the national debt can be used as money, assume that $M_t = D_t$ (so that $B_t = 0$). The central bank conducts monetary policy by choosing R_t .

As usual, I want to restrict attention to stationary allocations (i.e., a situation that may be expected to prevail in the long-run). To this end, assume that the Fed pegs a nominal interest rate R . In a stationary state, the stock of nominal debt grows (or shrinks) at a constant rate $\mu \equiv D_t/D_{t-1}$. Since $D_t = M_t$, we can use these restrictions to rewrite (1) as follows,

$$G_t - T_t = \left[1 - \frac{R}{\mu}\right] M_t \quad (3)$$

Define $x_t \equiv (G_t - T_t)/(N_t p_t)$. By stationarity, $x_t = x$ for all t . For simplicity, assume a constant population, so that $N_t = N$ for all t . Now, rewrite (3) as follows,

$$x = \left[1 - \frac{R}{\mu}\right] \left(\frac{M_t}{N p_t}\right) \quad (4)$$

2.3 Market-clearing

Because there is no investment and because this economy is closed, national saving is necessarily equal to zero (i.e., the entire RGDP is consumed). Our equilibrium concept is that, for a given level of real income (the RGDP is fixed at Ny in this model), the real interest adjusts to ensure that *desired*

⁴If preferences (2) were non-linear in c_2 , then the expected future lump-sum tax would have implications for desired saving for the young.

national saving is equal to zero at each and every date; i.e.,

$$Ns^D(r; \beta) - \frac{M_t}{p_t} = 0 \quad (5)$$

for all t . Condition (5) implies that in a stationary equilibrium, the price-level p_t must grow at the same rate as the nominal debt, M_t . That is, $\Pi = \mu$, so that $r = R/\mu$. Note that because μ is under the control of the fiscal authority, the long-run “money” growth rate in this economy is determined by fiscal policy.⁵

2.4 Collecting restrictions

Combine (5) with the right-hand-side of (4) and define

$$S(r; \beta) \equiv (1 - r)s^D(r; \beta) \quad (6)$$

where, recall, $r = R/\mu$ and β is a parameter that shifts the demand for “real money balances.” A (stationary) equilibrium consists of a pair (x, r) satisfying,

$$x = S(r; \beta) \quad (7)$$

For a given (r, β) , the equilibrium price-level can be recovered from the market-clearing condition (5),

$$p_t = \left[\frac{M_t}{Ns^D(r; \beta)} \right] \quad (8)$$

2.5 Laffer curve

Let us begin by investigating the properties of the bond-seigniorage function $S(r; \beta)$ as defined in (6). The first thing to notice is that for $r = 1$, we have $S(1; \beta) = 0$.

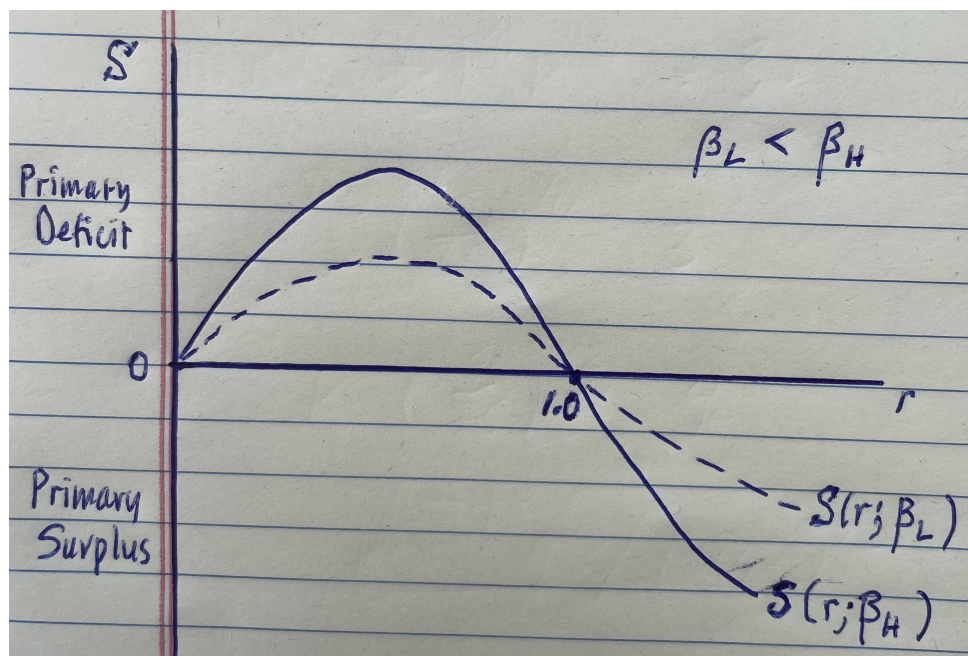
⁵This conclusion is not overturned if we instead adopt a conventional Monetarist assumption that the price-level is determined by the component of the national debt under central bank control (i.e., the monetary base). In a stationary equilibrium, the ratio of base money to bonds must remain constant. This implies that the monetary base must grow at the same rate as the total national debt.

Second, note that $S(r; \beta)$ is strictly decreasing in r for all $r > 1$. The intuition here is that if $R > \mu$, then “money-printing” is not sufficient to pay the interest expense of the debt. The government would have to, in this case, have to run primary budget surpluses (negative seigniorage) to finance the interest expense of the debt.

Third, note that $S(r; \beta) > 0$ for r over the range $[0, 1]$. Moreover, $S(r; \beta)$ has the “Laffer curve” property for r over this range of real interest rates. For very low values of r (e.g., very high rates of money growth μ), the government might actually increase bond-seigniorage revenue by *increasing* r (e.g., *lowering* the money growth rate μ). Beyond some critical (bond-seigniorage maximizing) r , the function $S(r; \beta) > 0$ begins to decline in r until $S(1; \beta)$.

Let us next consider how the bond-seigniorage function $S(r; \beta)$ depends on β , the parameter that indexes the demand for real bond balances. Recall from [A1] that $s^D(r; \beta)$ is increasing in β . It follows that $S(r; \beta)$ is increasing in β when $r < 1$ and is decreasing in β when $r > 1$. Figure 7.1 plots the function $S(r; \beta)$ for two values of β .

Figure 7.1



In what follows, I will assume that the economy operates somewhere to the right of the peak of the Laffer curve in Figure 7.1.

2.6 Policy coordination

Ricardian Fiscal Regime: A policy regime in which the monetary authority chooses R and the fiscal authority passively accommodates monetary policy through its choice of x (for a given μ) in a manner that satisfies the government budget constraint (7).

Non-Ricardian Fiscal Regime: A policy regime in which the fiscal authority chooses x and permits μ and R adjust passively to satisfy the government budget constraint (7).

3 Using the model

The model is best suited to studying the “long-run” effects of a given (permanent) change in parameters. But we can also think about how the transition might realistically look like in moving from one long-run situation to another.

3.1 An increase in the policy interest rate

Let us consider the effect of increasing the policy interest rate R at some arbitrary date t (and keeping the new interest rate elevated forever). The initial equilibrium must satisfy conditions (7) and (8). In a Ricardian regime, the fiscal authority lets x adjust, keeping μ fixed. In a Non-Ricardian regime, the fiscal authority lets μ adjust, keeping x fixed.

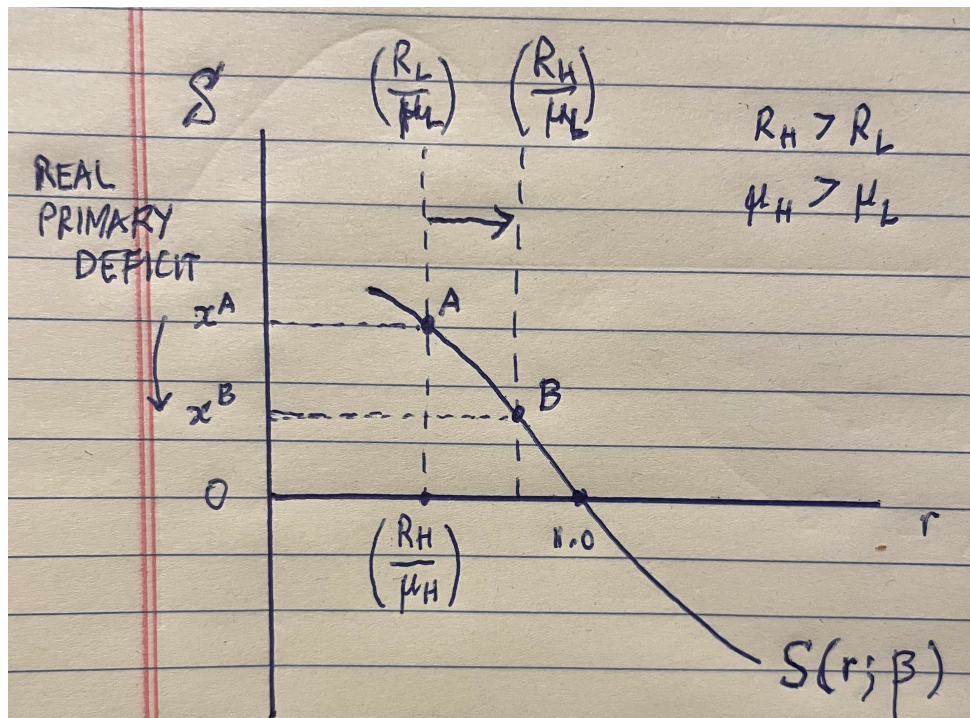
3.1.1 Ricardian fiscal regime

Let us consider the Ricardian regime first. An increase in R implies an increase in the real rate of interest, r . This has the effect of increasing the *real* demand for government debt, s^D . Because the supply of debt M_t is fixed at date t , equation (8) tells us that the price-level p_t must decline at date t . A decline in the price-level is the market’s way of making the real supply of

debt M_t/p_t increase in a manner that realigns supply with an elevated real demand for debt.⁶

Because an increase in the policy rate is referred to as “tightening monetary policy,” the model is consistent with the notion of tightening monetary policy to lower the rate of inflation. Note that the monetary authority is able to increase the real rate of interest here only because fiscal policy is Ricardian. That is, the resulting increase in the interest expense of the debt is met with a combination of real cuts to spending and/or higher taxes, instead of an increased pace of treasury-security issuance. That is, the real primary deficit x declines while μ remains unchanged. In Figure 7.2, the economy moves from point A to point B.

Figure 7.2



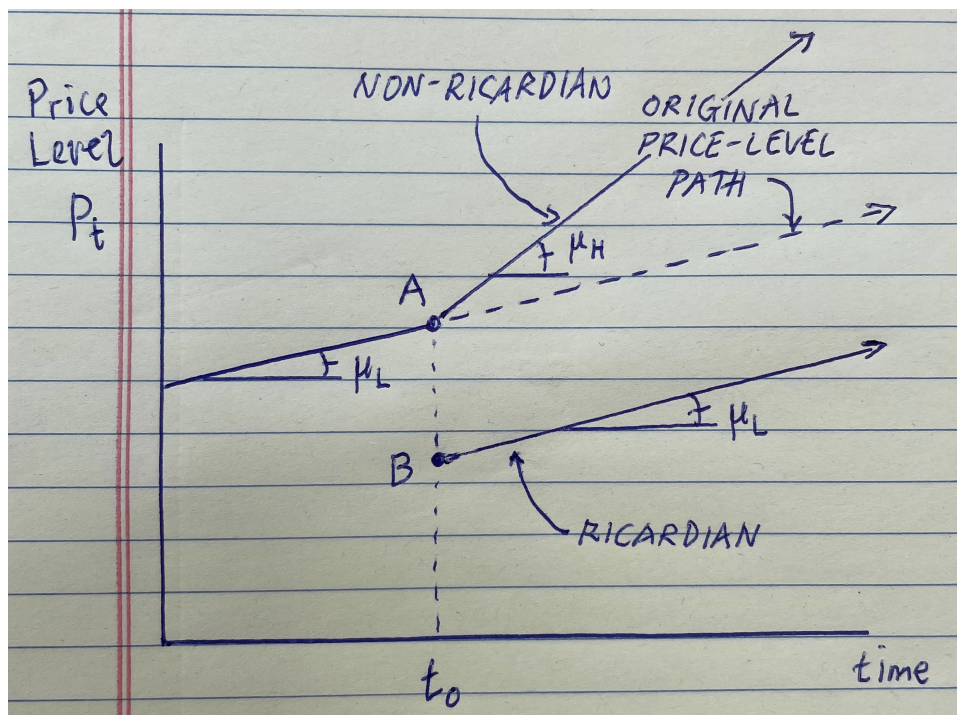
Note that because the long-run rate of inflation is determined by μ , and because μ remains unchanged in a Ricardian fiscal regime, it follows that

⁶In reality, adjustments could take place along a variety of dimensions, including the level of RGDP. We need to keep in mind that the model here assumes a constant RGDP.

the disinflation induced by a *permanent* tightening of monetary policy is *transitory*. This is consistent with a permanently lower price-level (see Figure 7.3). Because the RGDP is unaffected in this economy, a permanently lower price-level implies a permanently lower NGDP and a permanently higher debt-to-GDP ratio.

Proposition 7.1 When the fiscal regime is Ricardian, an increase in the policy interest rate (a tightening of monetary policy) leads to a *transitory* disinflation (a *permanent* decline in the price-level) and permanent increase in the debt-to-GDP ratio.

Figure 7.3



Does it not seem odd that a smaller real primary deficit is associated with a larger debt-to-GDP ratio?

Note that in this thought experiment, the total nominal deficit $M_t - M_{t-1}$ continues to grow at rate μ . Recall the government budget constraint,

$$X_t + (R_{t-1} - 1)M_{t-1} = M_t - M_{t-1} \quad (9)$$

where $X_t \equiv G_t - T_t$. In the period in which the policy rate is increased ($t = t_0$ in Figure 7.3), equation (9) remains unaffected. At date $t = t_0$, the price-level declines (A to B in Figure 7.3). Since the NGDP in this economy is given by $Y_t = p_t N y$, it follows that the NGDP declines as monetary policy is tightened. Because the total stock of nominal debt remains the same at M_t , it follows that the debt-to-GDP ratio rises.

The total deficit-to-GDP ratio must also rise, but in periods following the increase in R , the higher interest expense of the debt must cut into the primary deficit. Note that all that is needed here is a one-time cut in spending G_t and/or a one-time increase in taxes T_t . After the cut in the nominal primary deficit, spending and tax revenue can both continue to grow at rate μ .

3.1.2 Non-Ricardian fiscal regime

Let us now consider the effect of a tighter monetary policy when the fiscal regime is Non-Ricardian. Once again, the initial equilibrium must satisfy conditions (7) and (8). In a Non-Ricardian regime, the fiscal authority targets the real primary deficit x and lets μ adjust to satisfy the government budget constraint. Because the left-hand-side of (7) remains fixed, so must the right-hand-side of (7). Since the right-hand-side of (7) now entails a higher nominal interest rate, it follows that the pace of treasury-security issuance must increase one-for-one with the nominal interest rate.

The result of this thought experiment is depicted In Figure 7.2. The initial equilibrium position A remains in place following the tightening of monetary policy. The *real* interest rate remains unchanged, i.e., $(R_L/\mu_L) = (R_H/\mu_H)$. Because the real interest remains unchanged and because M_t is given at $t = t_0$, the price-level does not react in the period monetary policy is tightened. However, because the supply of nominal debt subsequently grows at an accelerated rate (which is necessary to pay for the higher interest expense of the debt), the rate of inflation increases (see Figure 7.3).

3.1.3 Discussion

It is conventional wisdom that an independent central bank can determine the long-run rate of inflation. Certainly, the Federal Reserve seems to believe this. Consider, for example, this statement from the Federal Open Market Committee’s “Statement on Longer Run Goals and Monetary Policy Strategy,” which states:⁷

“The inflation rate over the longer run is determined primarily by monetary policy, and hence the Committee has the ability to specify a longer-run goal for inflation.”

Suppose inflation has been running at 10% for years and that the central bank wants to lower the long-run inflation rate to 5%. How would it go about achieving this goal? If fiscal policy is Ricardian, a permanent increase in the interest rate will lower the price-level relative to where it would have been. Over this transition period, the economy will experience a lower rate of inflation. But if nothing else is done, the inflation rate is predicted to return to its original level. In order for the central bank to keep inflation low over time, it would have to keep increasing the real rate of interest every time the disinflationary pressure dissipated. This sounds like an unsustainable policy.⁸ At the end of the day, what is needed is fiscal reform that implies a lower μ . Unfortunately for our independent central bank, μ is under the control of the fiscal authority. A central bank cannot unilaterally determine the long-run rate of inflation.

The situation is even stranger in when fiscal policy is Non-Ricardian. The conventional wisdom of tightening monetary policy through increasing the policy rate actually backfires, generating an even higher rate of inflation. This possibility was first raised by Sargent and Wallace (1981), which they described as “Unpleasant Monetarist Arithmetic.”⁹ Loyo (1999) argues that this is exactly what explains the Brazilian experience with inflation in the

⁷See [https:// www.federalreserve.gov/ monetarypolicy/ files/fomc_longerrungoals.pdf](https://www.federalreserve.gov/monetarypolicy/files/fomc_longerrungoals.pdf)

⁸In reality, it would drive the economy into an ever-deeper recession. It may even violate certain central bank mandates. The Fed’s third mandate, for example, is “moderate long-term interest rates.”

⁹Sargent, Thomas and Neil Wallace (1981). “Some Unpleasant Monetarist Arithmetic,” *Federal Reserve Bank of Minneapolis Review*, Fall, pp. 1-15.

late 1970s and early 1980s.¹⁰ Interestingly, the central bank potentially does have greater control over the long-run rate of inflation in a Non-Ricardian regime. The prescription for lower inflation in this case, however, is to *lower* the policy rate. The effect of lowering the interest induces the Treasury to issue nominal debt at a slower pace, slowing the flow of outside assets that the private sector must absorb.

I suspect that some readers may object that I am assuming that the fiscal authority has fully control over the de facto money supply, M_t . What if we made the more traditional “Monetarist” assumptions that: (i) the central bank controls M_t ; and that (ii) the price-level depends on M_t and not on nominal debt D_t . We continue to assume (realistically) the monetary authority is not permitted to make transfers; it can only purchase (or sell) government securities. Thus, the fiscal authority still controls the total debt $D_t = M_t + B_t$ with the monetary authority determining the composition of the debt between M_t and B_t .

Next, consider our OLG model above but where M_t is distinct from B_t (so, money and bonds are not perfect substitutes). In this model, the price-level will be proportional to M_t and the real interest rate r_t is increasing in the bond-to-money ratio, B_t/M_t . In other words, we can think of monetary policy either as determining the ratio B_t/M_t or as determining r_t and letting the B_t/M_t adjust to where it needs to be to hit the target r_t .

In this Monetarist model, the central bank obviously has the capacity to determine the inflation rate. But this was also true of our model above. The question is whether the central bank can *unilaterally* determine the long-run rate of inflation by controlling the (narrow) money supply.

The answer turns out to be no. Consider a central bank that holds the supply of money fixed, $M_t = M$. Next, assume that the fiscal authority permits the supply of nominal Treasury debt to grow indefinitely at rate μ . It follows that the B_t/M ratio rises to infinity in the long-run, taking the real rate of interest with it. This is not likely to be a sustainable path. In a sustainable (i.e., stationary) equilibrium, the real interest rate and the bond-to-money ratio need to be constant. But if B_t/M_t is a constant, then M_t must grow at the same rate as B_t . The latter object is under fiscal control. Once again, we see that it is fiscal policy that ultimately determines the long-run

¹⁰Loyo, Eduardo (1999). “Tight Money Policy on the Loose: A Fiscalist Hyperinflation.” Unpublished manuscript. <http://sims.princeton.edu/yftp/Loyo/LoyoTightLoose.pdf>

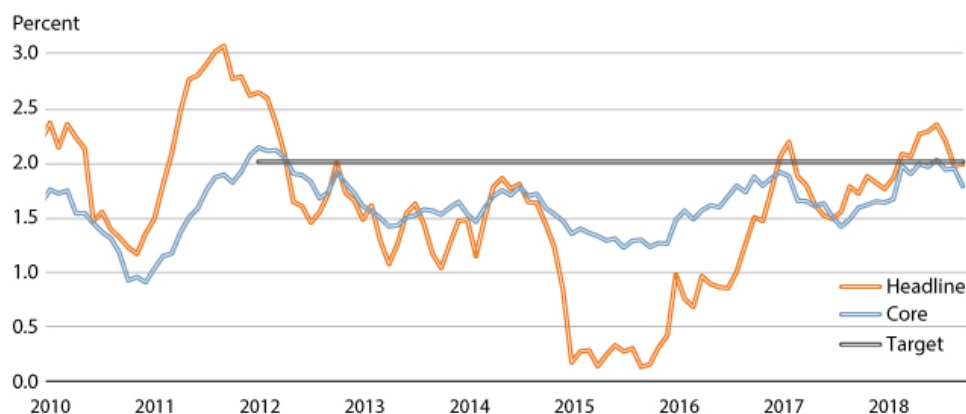
rate of inflation.

Having said this, our model does not tell us how long the “short-run” might last. In reality, the “short-run” might entail several years. If so, then our model is consistent with the notion of a central bank determining (or at least, influencing) the rate of inflation over some number of years without fiscal support (the early 1980s in the United States comes to mind). Nevertheless, the idea of determining a long-run inflation target without fiscal support seems a rather odd practice. In some countries, like Canada, the government and the central bank jointly determine the central bank’s monetary policy framework and inflation target. This joint endeavor recognizes the importance of fiscal policy in supporting the goals of monetary policy.

3.2 Understanding lowflation

“Lowflation” refers to a rate of inflation that persistently undershoots a target inflation rate. The Federal Reserve adopted a two percent inflation target in 2012. Usually, inflation overshoots the intended target so that central banks are almost always trying to lower inflation. But in the decade that followed the 2008-09 financial crisis and Great Recession, inflation remained below target in many countries, leaving central banks in an unfamiliar situation. Figure 7.4 plots the U.S. inflation experience from 2010-2018.

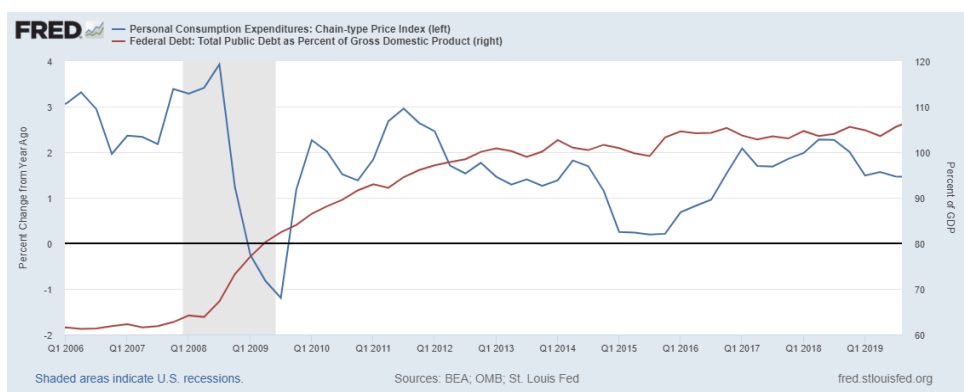
Figure 7.4



One way to interpret lowinflation is through the standard Monetarist lens of money demand growth outstripping money supply growth. We can index

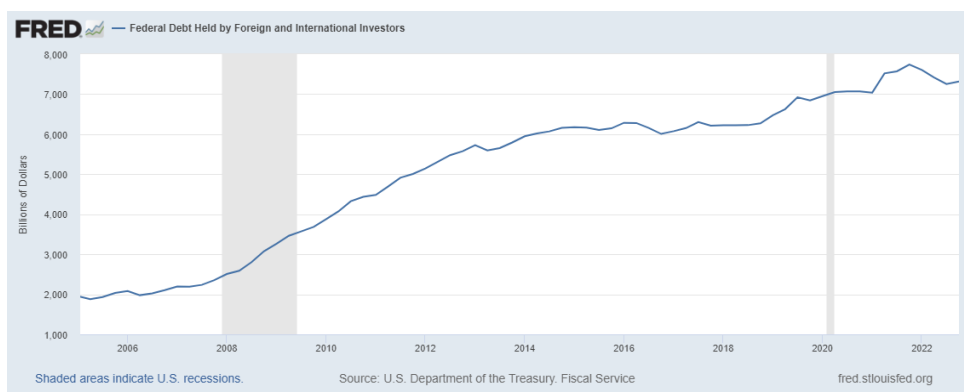
the strength of money demand by the parameter β . Recall [A1], which states that the demand for real money balances is increasing in β . It follows that the debt-to-GDP ratio is increasing in β as well. Figure 7.5 demonstrates that the U.S. debt-to-GDP ratio rose quite dramatically over this period of time. Let us assume then a sequence of β shocks $\beta_t < \beta_{t+1} < \beta_{t+2}$, etc., implying a corresponding $s^D(r; \beta_t) < s^D(r; \beta_{t+1}) < s^D(r; \beta_{t+2})$, and so on to a new higher level of β .

Figure 7.5



One contributing factor to the increase in the demand for USTs came from the role they played as a “flight-to-safety” security in the 2008-09 financial crisis. Another contributing factor was an increase in the global demand for USTs; see Figure 7.6.

Figure 7.6



Let us now reconsider our market-clearing condition (5), except with a time-varying β ,

$$Ns^D(r; \beta_t) - \frac{M_t}{p_t} = 0 \quad (10)$$

It follows that the equilibrium inflation rate over the period of transition from a low β to a high β is given by,

$$\Pi_{t+1} = \left(\frac{M_{t+1}}{M_t} \right) \left(\frac{s^D(r_t; \beta_t)}{s^D(r_{t+1}; \beta_{t+1})} \right) \quad (11)$$

Let me assume that the growth in the supply of debt $\mu_{t+1} \equiv M_{t+1}/M_t$ more or less tracked the increases in the demand for debt over the period in question. To the extent this is true, the inflation rate would have remained relatively stable over the period in question, which is broadly consistent with the behavior of core inflation in Figure 7.4. Once the transition is complete, μ_{t+1} reverts back to its normal level (see Figure 7.7) with $s^D(r, \beta)$ remaining elevated owing to a permanently higher β (which I assume is driving the permanently higher debt-to-GDP ratio in Figure 7.6).

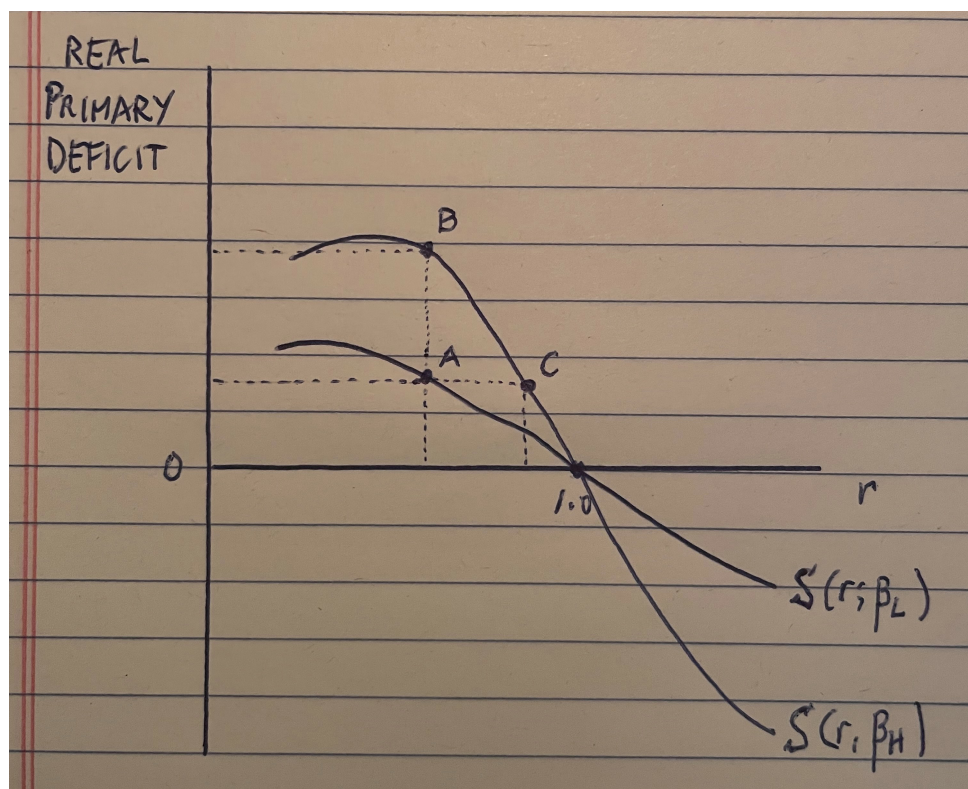
Figure 7.7



We can now refer to Figure 7.1 to depict the old and new long-run situations. Figure 7.8 depicts two bond-seigniorage functions for β_L (initial) and β_H (new), with $\beta_L < \beta_H$. The initial equilibrium is depicted as point A in Figure 7.8. Whether the new equilibrium is located at point B or point C depends on whether fiscal policy is Ricardian or Non-Ricardian. Note that the

nominal interest rate R did not change very much over this entire episode.

Figure 7.8



If fiscal policy is Ricardian, then μ should remain relatively unchanged prior to its initial long-run level (recall, we are permitting μ_t to deviate temporarily from these long-run levels). Because R did not change much (it was essentially stuck at its effective lower bound for most of the period), then our model predicts that the primary deficit as a ratio of GDP should have increased. Figure 6.7 in Chapter 6 shows that this is exactly what happened. Combined with the evidence presented in Figure 7.6, the conclusion is that the U.S. federal government increased its bond-seigniorage revenue without a higher inflation by extracting bond seigniorage from foreigners. Under a Ricardian fiscal policy, the economy moves from A to B in Figure 7.8

Suppose instead that fiscal policy is Non-Ricardian. For a Non-Ricardian fiscal policy, the economy moves from A to C. That is, all of the adjustment

takes place via μ instead of the primary deficit. Note that there is some evidence of this happening as well. In particular, note that the growth rate of the nominal debt near the end of the sample in Figure 7.7 is somewhat lower than it was prior to the financial crisis. So, the interpretation here is that an elevated demand for USTs gave rise to an expanded bond-seignorage tax base. This, in turn, meant that a given primary deficit to GDP ratio could be financed with a slower pace of treasury-security issuance. Essentially, the government is buying goods and services at lower prices than it otherwise would have paid (lower costs would have been passed through in the form of lower import prices as foreigners competed to buy USDs and USTs).

In reality, it seems plausible that the economy went from point A in Figure 7.7 to a point in between B and C. Perhaps fiscal policy was a little bit Ricardian and a little bit Non-Ricardian.

4 Conclusion

Unfortunately, we have no way of knowing for sure whether fiscal policy is better characterized as Ricardian or Non-Ricardian in real time. This is an especially important issue for monetary policy since, as we saw above, the anticipated effect of interest rate policy on inflation and the price-level depends critically on the fiscal regime. It is also important for monetary policy makers to ascertain the global demand for USTs.

In the post 2008-09 era, the demand for USTs grew immensely. The evidence for this could be seen in how bond yields plummeted even in the face of very large deficits. As bond yields approached their effective lower bound, the additional supply of nominal treasuries likely manifested themselves more through the price-level than through interest rates. The result was lowflation. Throughout this episode, the Fed promised to keep its policy rate low in an attempt to stimulate demand and raise inflation back to target. However, to the extent that fiscal policy was Non-Ricardian (and there is some evidence to suggest it was), the Fed should have perhaps been *raising* its policy rate for the purpose of returning inflation to its two-percent target.

The post Covid-19 era presents the Fed with a very different challenge. It is a challenge that central bankers are historically more familiar with (above target inflation). The Fed has been raising its policy rate very aggressively

in an attempt to lower the rate of inflation. This type of policy no doubt exerts disinflationary pressure through the standard economic channels. A more interesting question, however, is whether the Fed can get inflation to target and keep it there without fiscal support. Our analysis above suggests fiscal support is critical if the Fed is to achieve its mission.¹¹

Exercise 7.1 Use the analysis associated with Figure 7.2 to explain how an increase in the real primary budget deficit can lead to higher inflation for a given nominal interest rate. How should monetary policy set its policy interest rate to combat higher inflation in this scenario. Explain.

¹¹If the fiscal support is in place, then inflation should come down on its own in time. The Fed can nevertheless make the disinflation proceed more rapidly than it otherwise would by raising its policy rate.